

James Sypherd

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PROFESSIONAL SUMMARY

Game designer and software engineer with hands-on experience building gameplay prototypes, player-facing systems, and motion capture driven interactive experiences. Strong instincts for class identity, reward loops, combat readability, and player motivation, backed by years of MMO and tabletop systems analysis and cross-functional development work. Experience across Unity, Unreal Engine 5, UEFN, Roblox, and technical gameplay prototyping, with expertise in combat systems and progression design.

EXPERIENCE

Endless Games and Learning Lab, Arizona State University

Software Engineer / Game Developer December 2025 – Present

- Develop gameplay systems and prototype features for educational and entertainment-focused interactive projects across Unity, Unreal Engine 5, UEFN, and Roblox.
- Implement player-facing mechanics, interaction systems, and rapid gameplay prototypes in support of combat, progression, and experiential design goals.
- Support motion capture and .c3d data workflows for gameplay, animation, and interactive media systems.
- Contribute to Unreal Engine 5 development work tied to The Blacksmith, including gameplay prototyping and mechanic implementation.
- Translate design ideas into playable mechanics, tools, and technical workflows while collaborating across design, technical, and research contexts.

Arizona State University

Professor of Practice at the GAME School/GTA, AME 294: Martial Arts and Motion Capture Spring 2025 – Present

- Teach a 3-credit university course focused on motion capture, movement analysis, and real-time 3D visualization.
- Help students translate physical motion into readable interactive systems using OptiTrack workflows.
- Create instructional materials and technical documentation for creative technology pipelines and lab work.

Arizona State University Motion Capture Research

Graduate Researcher, Geometric Media Laboratory Fall 2024 – Spring 2025

- Built MATLAB tools to import motion capture data and perform inverse kinematics analysis on OptiTrack recordings.
- Designed capture and analysis workflows for studying timing, movement quality, and biomechanical variation.
- Developed practical expertise turning physical performance data into usable digital systems and visualizations.
- Earned the Endless Games and Learning Lab microcredential for Motion Capture for Martial Arts.

Hunting Grounds Board Game Development Team

Lead Game Designer Spring 2023 – Summer 2023

- Led design of a modular tabletop game centered on procedural learning through play.
- Built a prototype with 200+ cards, class-specific interactions, progression structures, and role-based mechanics.
- Designed itemization, rewards, and player guidance systems to strengthen replayability and class differentiation.

GAME DESIGN EXPERIENCE

World of Warcraft / Final Fantasy XIV

MMO Systems Analysis 2016 – Present

- Studied class design, talent structures, progression systems, encounter readability, and long-term reward loops across major expansion eras.
- Regularly analyze how class fantasy, accessibility, pacing, and combat feel shape player satisfaction and expression.
- Bring deep familiarity with MMO combat expectations, player perception, and the tradeoffs behind tuning and systems changes.

Dungeons & Dragons / Pathfinder

Dungeon Master and Systems Designer 2017 – 2025

- Ran weekly homebrew campaigns for 6 to 8 players with custom encounters, reward systems, and class interaction design.
- Created custom rules, classes, items, and conversion systems between D&D 5e and Pathfinder 2e.
- Balanced multiple player needs while preserving fantasy, pacing, and clarity in cooperative combat experiences.

Daggerheart

Playtester Spring 2024 – Summer 2024

- Selected for Daggerheart playtesting and contributed structured analysis grounded in play experience and mechanic breakdown.
- Practiced systematic feedback, combat critique, and written reporting on gameplay systems.

EDUCATION

Arizona State University

M.S., Biomedical Engineering, May 2025 | GPA: 3.50

B.S.E., Biomedical Engineering, May 2024 | GPA: 3.60

SKILLS

Design: Systems design, combat analysis, reward loops, class interactions, encounter design, progression design, playtesting, technical documentation

Tools: Unreal Engine 5, UEFN, Unity, C#, C++, GitHub, MATLAB, Excel, OptiTrack, motion capture pipelines, .c3d workflows, prototyping tools

Strengths: Cross-functional collaboration, systems thinking, data analysis, teaching, presentation, iterative problem solving